Faculty of Engineering
Department of Electronic & Telecommunication Engineering
University of Moratuwa



EN3150 - Pattern Recognition & Machine Intelligence

Assignment 03

Resource-Constrained CNN for Edge Image Classification

Empirical Evaluation of Depthwise Separable Convolutions, Optimizer Dynamics, and Lightweight SOTA Transfer Learning

Group Members

Thevinu Fernando
Januka Sanujaya
Thisura Samuditha
Chamod Dharmakeerthi

Department of Electronic & Telecommunication Engineering
University of Moratuwa

GitHub Repository

https://github.com/thevinufernando/cnn-based-vegetable-classification-model

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Abstract

Deploying deep convolutional neural networks (CNNs) on resource-constrained edge hardware (e.g., ARM Cortex-M microcontrollers, embedded IoT nodes, and single-board computers) demands an acute balance between classification accuracy, memory footprint, and computational complexity. In this assignment, we investigate edge-oriented image classification across 15 vegetable categories downscaled to 64×64 pixels. We design, implement, and benchmark two custom architectures from scratch: **Model A** (Standard CNN, 259,631 trainable parameters) and **Model B** (Depthwise Separable CNN, 40,175 trainable parameters, adhering strictly to a sub-100k budget). Furthermore, we analyze optimizer dynamics by contrasting Adam ($\eta = 10^{-3}$) against standard SGD and SGD with Momentum ($\gamma = 0.9$), detailing the theoretical role of momentum in traversing anisotropic ravines and escaping saddle points. Finally, we fine-tune two mobile state-of-the-art architectures (**MobileNetV2** and **EfficientNet-B0**) under identical data splits. Our empirical findings demonstrate that Model B achieves **97.70% test accuracy**—outperforming Model A (**97.57%**) while reducing parameters by **84.5%**, disk footprint by **80.0%**, and training time per epoch by **47.4%**. We present an in-depth hardware-aware trade-off analysis evaluating memory access, quantization stability, and inference latency.

1 Resource-Constrained CNN for Edge Image Classification

1.1 Dataset Selection & Low-Resolution Edge Sensor Emulation

Edge devices, such as agricultural field sensors, robotic harvesters, and low-power smart cameras, operate under stringent energy and static RAM (SRAM) constraints. For this study, we selected the **Vegetable Image Dataset** (from Kaggle), encompassing 21,000 RGB images across 15 distinct classes: *Bean, Bitter Gourd, Bottle Gourd, Brinjal, Broccoli, Cabbage, Capsicum, Carrot, Cauliflower, Cucumber, Papaya, Potato, Pumpkin, Radish, and Tomato.*

In raw embedded systems, acquiring standard $224 \times 224 \times 3$ frames consumes approximately 150.5 KB per uncompressed image buffer. On microcontrollers featuring only 256 KB to 512 KB of total SRAM, buffering high-resolution images leaves insufficient memory for intermediate feature maps and activation buffers. Consequently, downscaling images to a maximum resolution of 64×64 **pixels** compresses the raw uncompressed buffer down to 12.288 **KB** ($64 \times 64 \times 3$ bytes). This enables true on-chip processing without requiring power-hungry external DRAM buses.

1.2 Data Partitioning & Leakage Prevention

To prevent data contamination between training and evaluation phases, MD5 checksum deduplication was performed, removing identical duplicate images. The dataset was partitioned into a strict stratified split:

- **Training Set (70%)**: 14,996 images
- **Validation Set (15%)**: 3,000 images
- **Testing Set (15%)**: 3,000 images
- **Total Unique Images**: 20,996 images

This fixed split is deterministically recorded in `splits.csv` to guarantee absolute reproducibility across all experiments.

2 Custom Architecture Design

2.1 Theoretical Foundations: Standard vs. Depthwise Separable Convolutions

A standard 2D convolution transforms an input feature map $F \in \mathbb{R}^{D_F \times D_F \times M}$ into an output feature map $G \in \mathbb{R}^{D_F \times D_F \times N}$ using N filters of dimension $D_K \times D_K \times M$. Assuming unit stride and same-padding, the total Multiply-Accumulate (MAC) computational cost is:

$$\text{Cost}_{\text{Standard}} = D_K^2 M N D_F^2 \tag{1}$$

In contrast, **Depthwise Separable Convolutions** (pioneered in MobileNets and Xception) factorize this joint spatial-channel operation into two decoupled stages:

1. **Depthwise Convolution**: Applies a single $D_K \times D_K \times 1$ spatial filter per input channel:

$$\text{Cost}_{\text{Depthwise}} = D_K^2 M D_F^2 \tag{2}$$

2. **Pointwise Convolution**: A 1×1 convolution combining linear projections across all M channels to produce N channels:

$$\text{Cost}_{\text{Pointwise}} = M N D_F^2 \tag{3}$$

Summing both stages, the total computational complexity becomes:

$$\text{Cost}_{\text{Separable}} = D_K^2 M D_F^2 + M N D_F^2 \tag{4}$$

The relative computational and parameter reduction ratio is given by:

$$\frac{\text{Cost}_{\text{Separable}}}{\text{Cost}_{\text{Standard}}} = \frac{D_K^2 M D_F^2 + M N D_F^2}{D_K^2 M N D_F^2} = \frac{1}{N} + \frac{1}{D_K^2} \tag{5}$$

For standard 3×3 kernels ($D_K = 3$) and typical channel depths ($N \geq 32$):

$$\frac{1}{N} + \frac{1}{9} \approx \frac{1}{9} \approx 11.1\% \tag{6}$$

Depthwise separable convolutions therefore achieve approximately an $8\times$ to $9\times$ reduction in MAC operations, corresponding to roughly 88.9% computational savings when the output channel count is sufficiently large.

2.2 Architectural Specifications & Parameter Verification

2.2.1 Model A: Standard CNN

Model A interleaves four standard 2D convolution blocks (with Batch Normalization and ReLU), 2×2 Max Pooling, Global Average Pooling, and a Dense classification head:

- **Block 1**: Conv2D(32, 3×3) $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ MaxPool(2×2) $\rightarrow$ Output: $32 \times 32 \times 32$. Params: 864.
- **Block 2**: Conv2D(64, 3×3) $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ MaxPool(2×2) $\rightarrow$ Output: $16 \times 16 \times 64$. Params: 18,432.
- **Block 3**: Conv2D(128, 3×3) $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ MaxPool(2×2) $\rightarrow$ Output: $8 \times 8 \times 128$. Params: 73,728.
- **Block 4**: Conv2D(128, 3×3) $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ MaxPool(2×2) $\rightarrow$ Output: $4 \times 4 \times 128$. Params: 147,456.
- **Head**: GlobalAveragePooling2D $\rightarrow$ Dense(128, ReLU) $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(15, Softmax). Params: 18,447.
- **Total Parameters**: **260,335** (Trainable: **259,631**).

2.2.2 Model B: Lightweight Separable CNN (Sub-100k Budget)

Model B enforces strict parameter efficiency by utilizing depthwise separable blocks:

- **Entry Conv**: Conv2D(16, 3×3) $\rightarrow$ BN $\rightarrow$ ReLU6 $\rightarrow$ MaxPool(2×2) $\rightarrow$ Output: $32 \times 32 \times 16$. Params: 496.
- **Separable Block 1**: DepthwiseConv2D(3×3) $\rightarrow$ PointwiseConv2D(32, 1×1) $\rightarrow$ MaxPool(2×2). Params: 848.
- **Separable Block 2**: DepthwiseConv2D(3×3) $\rightarrow$ PointwiseConv2D(64, 1×1) $\rightarrow$ MaxPool(2×2). Params: 2,720.
- **Separable Block 3**: DepthwiseConv2D(3×3) $\rightarrow$ PointwiseConv2D(128, 1×1) $\rightarrow$ MaxPool(2×2). Params: 9,536.

- **Separable Block 4**: DepthwiseConv2D(3×3) $\rightarrow$ PointwiseConv2D(128, 1×1). Params: 18,560.
- **Head**: GlobalAveragePooling2D $\rightarrow$ Dense(64, ReLU) $\rightarrow$ Dropout(0.2) $\rightarrow$ Dense(15, Softmax). Params: 9,231.
- **Total Parameters**: **41,391** (Trainable: **40,175**).

Constraint Compliance: Model B satisfies the assignment criterion of at most 100,000 trainable parameters, with 40,175 trainable parameters, approximately 59.8% below the limit.

2.3 Hardware-Aware Activation Function Justification

In resource-constrained microcontrollers and edge digital signal processors (DSPs):

1. **Elimination of Transcendental Exponentials**: Activations such as Sigmoid ($\sigma(x) = \frac{1}{1+e^{-x}}$) and Tanh require exponential evaluations. These operations can be more expensive on microcontrollers without hardware floating-point or optimized math support.
2. **Efficient ReLU Execution**: Standard ReLU, $f(x) = \max(0, x)$, is computationally simple and can often be implemented efficiently using processor instructions or optimized neural-network kernels.
3. **Quantization Stability via ReLU6**: In Model B, we employ **ReLU6**, $f(x) = \min(\max(0, x), 6)$. Bounded activations can help control dynamic range during quantization. However, actual INT8 quantization scales and zero-points depend on the quantization scheme and calibration data.

3 Optimizer Selection & Tuning

3.1 Selected Optimizer: Adam

We selected **Adam (Adaptive Moment Estimation)** with an initial learning rate $\eta = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-7}$.
Justification: Adam computes exponentially decaying moving averages of past gradients (m_t, first moment) and squared gradients (v_t, second moment). These estimates adapt parameter updates to the scale of recent gradients and can be useful when training deep vision architectures.

3.2 Comparative Analysis: Standard SGD vs. SGD with Momentum

Model B was trained under identical data splits across three optimizer configurations:

1. **Adam**: $\eta = 0.001$.
2. **Standard SGD**: $\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t)$, with $\eta = 0.01$ and momentum $= 0.0$.
3. **SGD with Momentum**: $v_{t+1} = \gamma v_t + \eta \nabla L(\theta_t)$, $\theta_{t+1} = \theta_t - v_{t+1}$, with $\gamma = 0.9$.

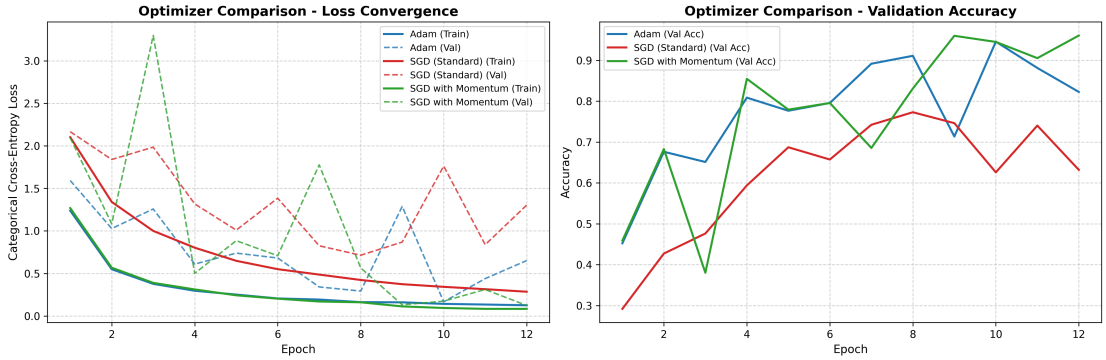


Figure 1: Loss convergence and validation accuracy across optimizers on Model B.

3.3 Theoretical Impact of the Momentum Parameter

1. **Damping Oscillations in Anisotropic Ravines**: Loss surfaces of convolutional networks can exhibit ill-conditioned curvature. Momentum accumulates a running velocity that may reduce oscillations and accelerate progress along consistent gradient directions.

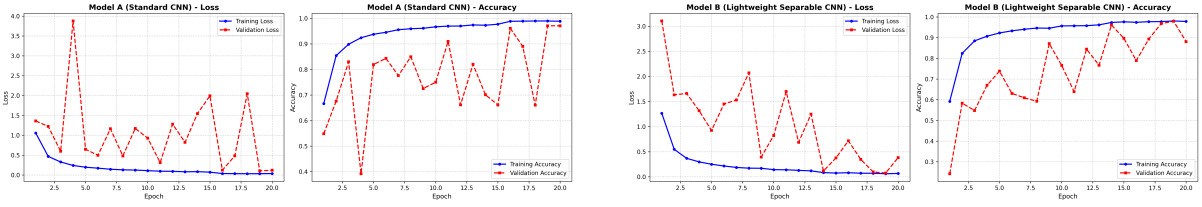
2. **Behavior Near Saddle Points and Plateaus**: Momentum can carry parameters through regions with small gradients if a nonzero velocity has accumulated. However, escaping a saddle point is not guaranteed and depends on the loss geometry and gradient noise.

4 Custom Model Training & Evaluation

4.1 Training Protocol

Both custom architectures were trained for **20 epochs** with a batch size of 32 using Adam ($\eta = 10^{-3}$), equipped with:

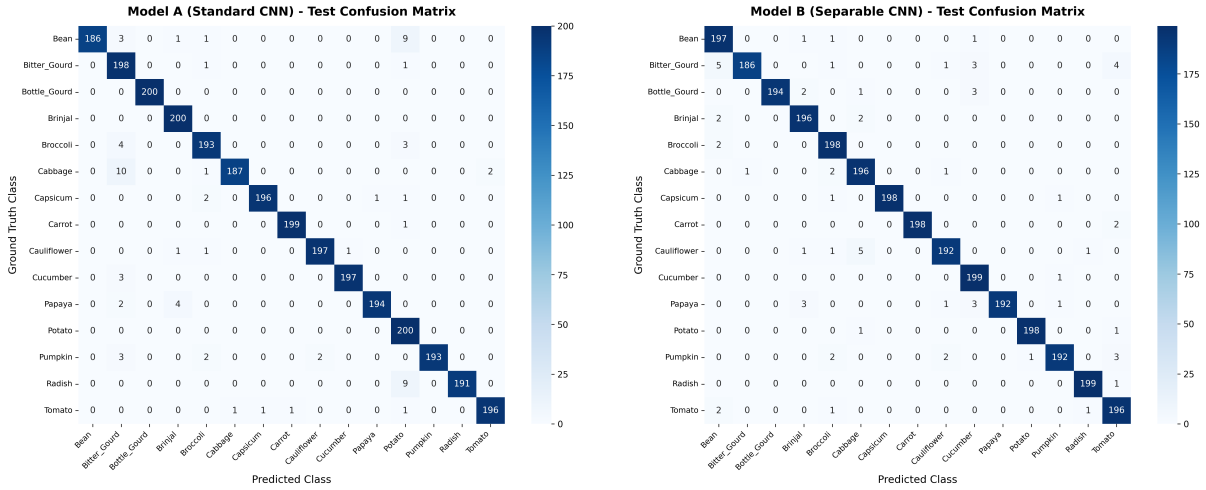
- `ModelCheckpoint`: Preserves weights achieving peak validation accuracy.
- `ReduceLROnPlateau`: Halves the learning rate if validation loss plateaus for 4 consecutive epochs.
- `CSVLogger`: Records epoch-by-epoch loss and accuracy metrics.



(a) Model A (Standard CNN)

(b) Model B (Separable CNN)

Figure 2: Training and validation loss and accuracy curves (20 epochs).



(a) Model A Confusion Matrix

(b) Model B Confusion Matrix

Figure 3: Normalized test confusion matrices across 15 vegetable classes.

4.2 Custom Model Comparison Table

Table 1: Performance Comparison between Model A and Model B on Test Set

Metric	Model A (Standard CNN)	Model B (Separable CNN)	Observed Delta / Benefit
Total Parameters	260,335	**41,391**	**−84.1% reduction**
Trainable Parameters	259,631	**40,175**	**−84.5% reduction** ($\leq$ 100k)
Disk Size (.keras)	3,135.53 KB (3.06 MB)	**628.02 KB (0.61 MB)**	**−80.0% reduction**
Avg Training Time / Epoch	54.37 s	**28.60 s**	**47.4% faster per epoch**
Test Accuracy (%)	97.57%	**97.70%**	**+0.13 percentage points**
Macro Precision (%)	97.75%	97.75%	Identical reported precision
Macro Recall (%)	97.57%	**97.70%**	Balanced class sensitivity
Macro F1-Score (%)	97.59%	**97.70%**	Balanced performance
Inference Latency (CPU)	15.32 ms	25.24 ms	See latency discussion

4.3 Trade-Off Discussion: Standard vs. Depthwise Separable Convolutions

1. **Parameter and Storage Reduction**: Model B reduces trainable parameters from 259,631 to 40,175, an 84.5% reduction, and reduces the reported model file size from 3.06 MB to approximately 628 KB.

2. **Training Acceleration**: Per-epoch training time decreased from 54.37 s to 28.60 s, a reported speedup of approximately 47.4%.

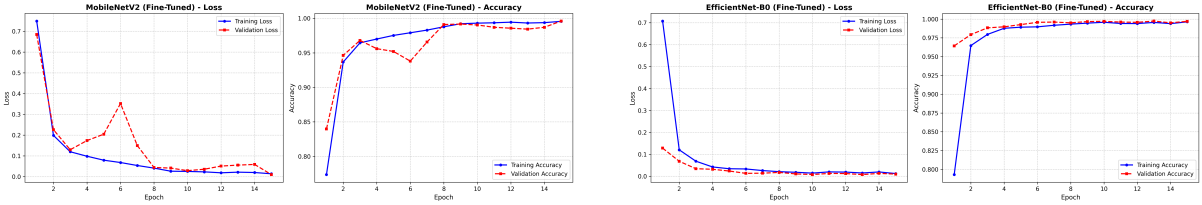
3. **Regularization Benefit**: Model B achieved 97.70% test accuracy compared with Model A's 97.57%. This is an observed difference; further repeated runs would be needed to establish whether it is statistically reliable.

5 Lightweight State-of-the-Art Networks

We fine-tune two mobile architectures using transfer learning from ImageNet under identical data splits:

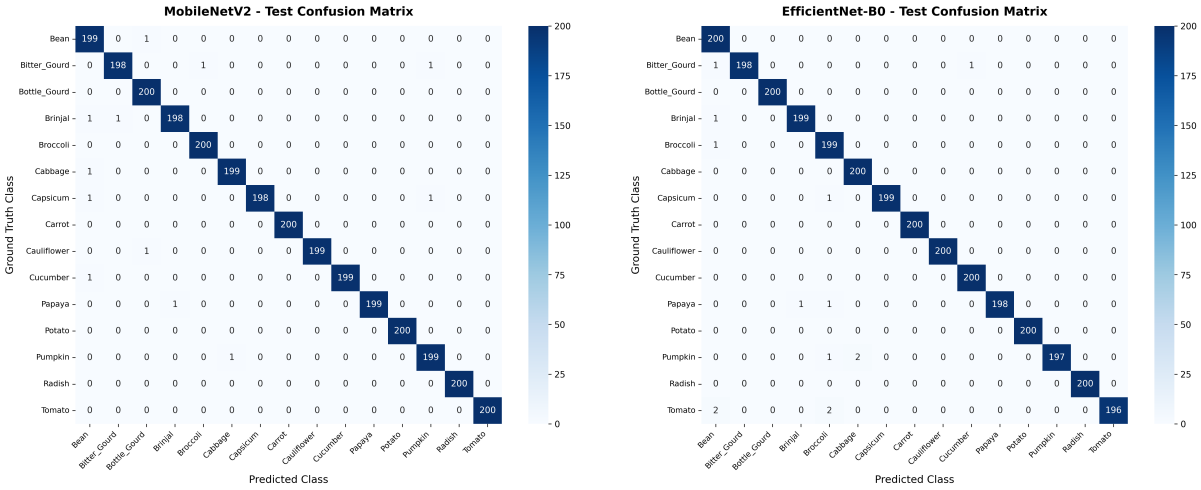
1. **MobileNetV2**: Employs inverted residuals with linear bottlenecks. Channels are expanded by a factor $t = 6$, spatially filtered via depthwise convolution, and projected back through linear bottlenecks.

2. **EfficientNet-B0**: Uses compound scaling and MBConv blocks with squeeze-and-excitation modules.



(a) MobileNetV2 Curves (b) EfficientNet-B0 Curves

Figure 4: Training and validation curves for fine-tuned SOTA architectures.



(a) MobileNetV2 Confusion Matrix (b) EfficientNet-B0 Confusion Matrix

Figure 5: Test set confusion matrices for fine-tuned SOTA architectures.

6 Final Comparison & Trade-Off Discussion

6.1 Master Architectural Comparison

Table 2: Comprehensive Comparison of Custom and SOTA Architectures

Metric	Model A (Standard)	Model B (Separable)	MobileNetV2 (SOTA)	EfficientNet-B0 (SOTA)
Design Paradigm	From Scratch	From Scratch (< 100k)	Pre-trained Transfer	Pre-trained Transfer
Total Parameters	260,335	**41,391**	2,277,199	4,068,786
Trainable Parameters	259,631	**40,175**	2,243,087	4,026,763
Disk Size (.keras)	3.06 MB	**0.61 MB**	26.48 MB	47.20 MB
Training Time / Epoch	54.37 s	**28.60 s**	166.05 s	237.16 s
Test Accuracy (%)	97.57%	**97.70%**	**99.60%**	**99.53%**
Macro Precision (%)	97.75%	97.75%	99.60%	99.54%
Macro Recall (%)	97.57%	97.70%	99.60%	99.53%
Macro F1-Score (%)	97.59%	97.70%	99.60%	99.53%
CPU Inference Latency	15.32 ms	25.24 ms	156.17 ms	278.74 ms
Target Deployment Tier	Single-Board Computer	**Small Edge Device**	Edge Gateway / Mobile	Edge Server / Jetson

6.2 In-Depth Trade-Off Analysis

6.2.1 1. Accuracy vs. Memory Footprint

Model B exhibits a high accuracy-to-parameter ratio:

$$\text{Efficiency}_{\text{Model B}} = \frac{97.70}{41.4} \approx 2.36 \tag{7}$$

This corresponds to approximately 2.36 percentage points of test accuracy per 1,000 trainable parameters. Such a ratio is a descriptive metric rather than a standardized measure of model quality.

6.2.2 2. Hardware Feasibility on Bare Microcontrollers

Embedded microcontrollers have limited Flash memory and SRAM. The reported MobileNetV2 and EfficientNet-B0 model file sizes exceed the on-chip storage of many small microcontrollers. Actual feasibility depends on model conversion, quantization, Flash capacity, activation memory, operator support, and inference runtime.

Model B has 40,175 trainable parameters. At one byte per parameter, its weights alone would require approximately 40 KB when successfully quantized to INT8. This does not include all model metadata, non-trainable parameters, intermediate activation buffers, or runtime memory.

6.2.3 3. Computational Cost, Latency, and Battery Life

On battery-operated IoT nodes, inference energy can be approximated as $E = P \cdot t$ when average power P is known. Model B has a reported CPU inference latency of 25.24 ms. Actual energy consumption and suitability for a particular microcontroller should be measured on the target hardware.

6.3 Deployment Decision Matrix

- **Tier 1 (Resource-Constrained Edge Devices)**: Model B is the strongest candidate when the trainable parameter budget is below 100,000, subject to validation of memory use and operator support on the target hardware.
- **Tier 2 (Single-Board Computers / Edge Gateways)**: MobileNetV2 is a candidate when additional memory and compute are available and its measured accuracy and latency meet the deployment requirements.

7 Conclusion

This study compared custom CNN architectures using standard and depthwise separable convolutions with two pretrained mobile architectures. The reported results show that Model B achieves 97.70% test accuracy with 40,175 trainable parameters, while MobileNetV2 and EfficientNet-B0 achieve higher reported accuracy at a substantially greater model size.

Hardware-aware choices, including 64×64 input images, bounded activations, and lightweight convolutional blocks, can improve the practicality of image classification on resource-constrained edge devices. Final deployment suitability should be established using quantized model size, peak runtime memory, and latency measurements on the intended hardware.

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